KILNDECK · MARKET AND VIABILITY

KilnDeck market and viability

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

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0.1	2026-10-28	Amish Chadha	First issue (kit 1.8 upgrade)

KilnDeck market and viability

KilnDeck is for the fixed-chimney and zigzag brick kilns of South Asia, where firemen still feed fuel on foot on the kiln top; nothing on the market gives them a safe place to stand, so the benchmark is the general work platforms a kiln owner could adapt and the motorised feeders that research programmes have tried. An open design matters because the users are poor, the kilns are many and scattered, and the fabricators who would build it are local workshops, not equipment makers. Prices and benchmarks are cited with the date they were checked. Method: STANDARDS section 23.

Table 1. Market segments.

Segment	Target users and region	Competitive benchmark (cited price)	Estimated cost per unit (USD)	Value proposition
Fixed-chimney bull's trench kilns	Firemen and owners in Bangladesh, Pakistan (Punjab), India and Nepal	No product found that carries a fireman over the trench. Nearest general platform: an aluminium double-width mobile scaffold tower, INR 7,500 a piece for an 18 ft tower (IndiaMART listing, checked 2026-10-25); it cannot span the trench or stand on the crust	2,787 (BOM, one deck and 2 x 12 m of track)	Feeding without standing on the crust, from equipment a local workshop builds
Clean-kiln and worker safety programmes	NGO, research and government programmes that train firemen	Motorised coal feeders piloted by Stanford and BUET (price not published; to assess)	2,787	A hand-powered safety measure that sits beside fuel-efficiency training
Zigzag kilns (after conversion)	The same owners, after kiln conversion	To assess (feeding pattern and wall geometry differ)	to assess	Possible with a different track layout; not designed

Cost notes: steel is costed at USD 1.30 per kilogram in the BOM, an estimate; an Indian listing for mild steel rectangular pipe was INR 53 per kilogram (IndiaMART, Anjani Tubes, Ghaziabad, checked 2026-10-25). A 100 mm, 20 mm round-groove gate wheel rated 400 kg sells at GBP 12.20 before VAT in the UK (F.H. Brundle, checked 2026-10-25); the BOM's USD 30 for a 500 kg wheel is a regional estimate to confirm (KND-DEC-001 item 3). Fabrication labour is not included.

Open route

Who builds it: a steel fabricator in a brick district, from the drawings, the build plan and the BOM, paid by the kiln owner or a programme. How it spreads: through clean-kiln programmes and kiln owners' associations that already run training, and through the open repository (CERN-OHL-S-2.0 hardware, MIT software), so any workshop can build, change and share it. What keeps it open: the strongly reciprocal licence, which requires changes to be shared, and a design that uses only stock steel and catalogue bearings, wheels and chain. KilnDeck is a non-commercial open engineering reference; no sale is planned.

Sources

- IndiaMART, JPR Ladders (Sunjet Access Solution), mobile scaffolding listings, https://m.indiamart.com/jprladders/mobile-scaffolding.html, accessed 2026-10-25.
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- F.H. Brundle, "IBFM 385100L 20mm Round Groove Wheel 100mm With Support Max Gate 400Kgs," GBP 12.20 ex VAT, https://www.fhbrundle.co.uk/products/0515W002__20mm_Round_Groove_Wheel_100mm_With, accessed 2026-10-25.
- Luby Lab, Stanford University, "Automatic Coal Feeder," https://lubylab.stanford.edu/automatic-coal-feeder (cited in KND-PRB-001).